

Question number	Answer	Additional guidance	Mark
2(a)	An explanation that includes two of the following points: - antibiotics used to treat impetigo and whooping cough because they are caused by bacteria (1) - antibiotics not always used to treat middle ear infections or sinus infections as they {can be caused by viruses / as they are not always caused by bacteria} (1) - antibiotics not used to treat multiple sclerosis or rheumatoid arthritis as they are not caused by bacteria (1)	NB Do not accept if clear confusion with antibodies ACCEPT because some of the bacteria are resistant to antibiotics IGNORE any ref to cause ACCEPT antibiotics used for bacterial infections if no other marks awarded	(2)

Question number	Answer	Mark
2(b)	The only correct answer is B. <i>A is incorrect because bactericidal antibiotics decrease the number of bacteria</i> <i>C is incorrect because antibiotics do not increase the number of bacteria</i> <i>D is incorrect because antibiotics do not increase the number of bacteria</i>	(1)

Question number	Answer	Additional guidance	Mark
2(c)(i)	An explanation that includes the following points: - to compete with (pathogenic) bacteria for {nutrients / named nutrient / space} (1) - so they {reduce / destroy / prevent the growth of} (pathogenic) bacteria (1)	IGNORE food ACCEPT produce {toxins / chemicals} ACCEPT so that they do not increase in number and cause disease	(2)

Question number	Answer	Additional guidance	Mark
2(c)(ii)	An explanation that includes the following points: (because gut flora contain) bacteria (1) - and antibiotics are not (generally) specific to one type of bacteria (1)	ACCEPT antibiotics {can affect different types of bacteria / can be broad spectrum}	(2)

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2(c)(iii)	accept a value between 9×10^6 and 1.2×10^7		(1)

Question number	Answer	Additional guidance	Mark
2(c)(iv)	An answer that includes the following points: • antibiotic P kills all but three types (1) • antibiotic Q only kills two types (1) • antibiotic R has no effect (on gut flora / types A and B) (1)	ACCEPT descriptions in terms of resistance and susceptibility but not immunity throughout ACCEPT correct references to bactericidal and bacteriostatic ACCEPT kills 4 types / kills type A but has limited effect on type B ACCEPT results in the presence of a new type / reduces the number of type A and type B	(3)

